

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
BSc (Hons) Computing|ITB|EHN|Year-1-19.1
DIPLOMA IN SOFTWARE ENGINEERING 19.1
COMPUTER TECHNOLOGY - NIB-SOC-COM-2-3-05

22nd August 2019, 09.00 am to 11.00 am

Answer ALL questions. (50 Marks)

Time: Two hours.

1.

- a) Describe the current conducting mechanism of a n-type semiconductor with a diagram.
(5 marks)
- b) Describe the ‘forward biased operating mode’ associated with a p-n junction diode.
(5 marks)
- c) Draw a schematic diagram of a full-wave rectifier circuit with smoothing capacitor and explain its operation using appropriate waveforms.
(10 marks)
- d) Figure 1.1 depicts a circuit containing a Ge diode. Also Figure 1.2 depicts the typical characteristics curves of a diode. The external voltage source is 9V.
(5 marks)

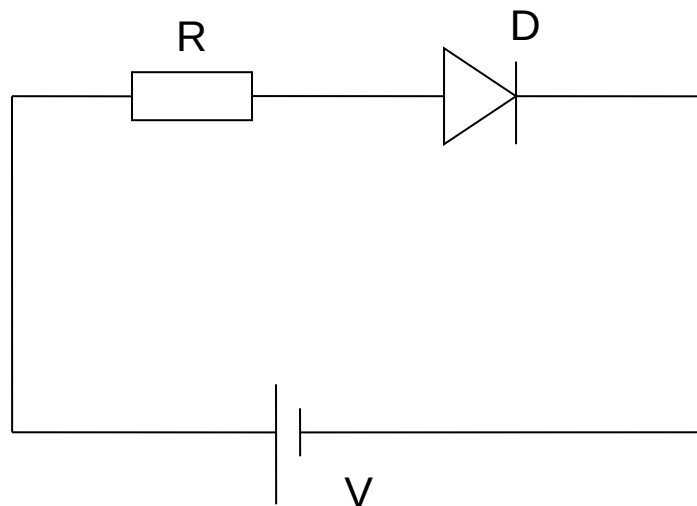


Figure 1.1

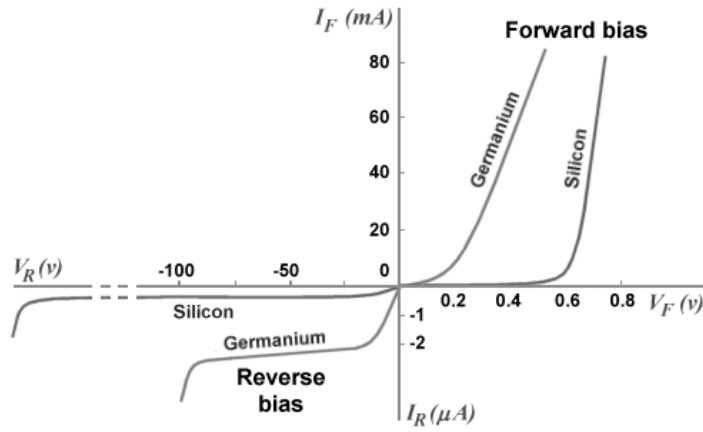


Figure 2.2

- Find the approximate values for peak inverse voltage and reverse saturation current.
- What is the forward voltage for forward current of 40mA?
- Assume that current passing through the circuit is 40mA, and then calculate voltage different across the resister and resistance of the resistor.

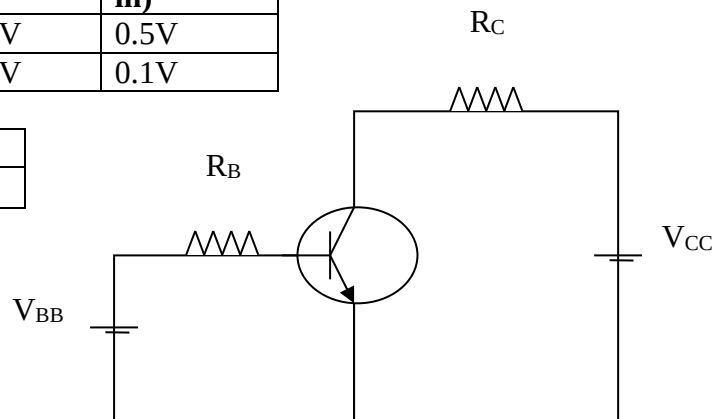
(25 Marks)

2.

- What are the two types of Bipolar Junction Transistors? Draw their circuit symbols. (5 marks)
- Explain the switched mode operation of a BJT by referring its output characteristic curve. (5 marks)
- Consider the following circuit and obtain the operating mode of the transistor. Hence calculate I_B , I_C and V_{CE} . Assume that the transistor is made out of Si. (5 marks)

Material	$V_{CE}(\text{sat})$	$V_{BE}(\text{sat})$	$V_{BE}(\text{act})$	$V_{BE}(\text{cut in})$
Si	0.2V	0.8V	0.7V	0.5V
Ge	0.1V	0.3V	0.2V	0.1V

β	R_B	R_C	V_{CC}	V_{BB}
50	200K Ω	1K Ω	12V	6V



- Consider the circuit in figure 2.1 and the characteristic curve of the JFET in figure 2.2. If $V_{DD} = 10V$, $V_{GG} = 0.9V$, $R_G = 1K\Omega$, $R_D = 8K\Omega$. (10 marks)

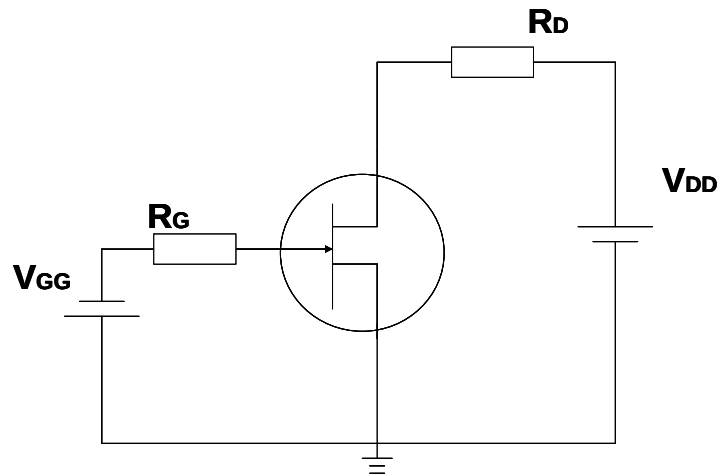


Figure 2.1

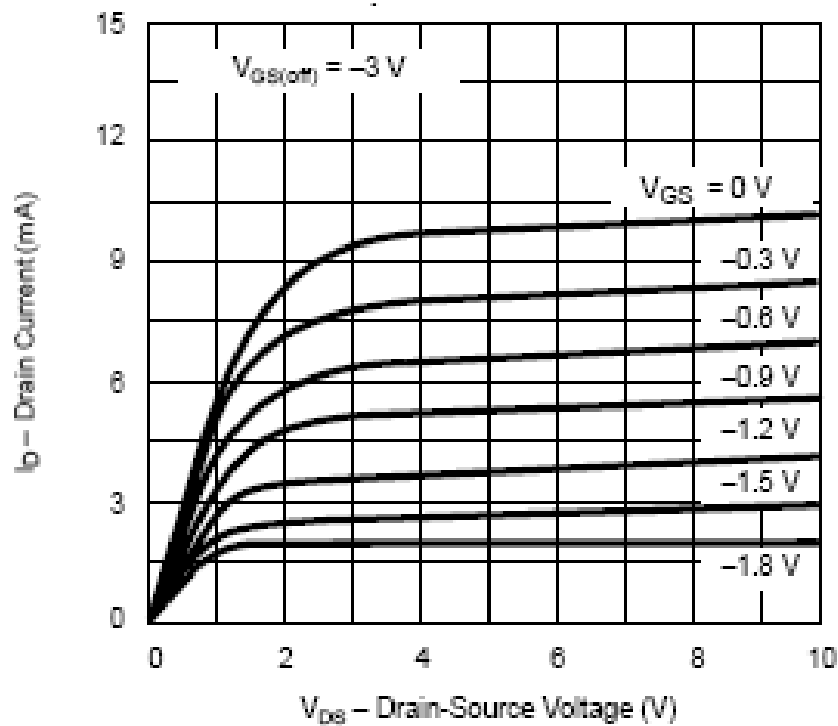


Figure 2.2

- What is the type of the given JFET?
- Indicate the three regions of the characteristic curve.
- Calculate Gate-Source voltage.
- Obtain an equation for the load line of the circuit and plot it over the given characteristic curve.
- Hence identify the operating mode of the JFET and find the coordinate of the operating point.

(25 Marks)